

Human digestion

THE DIGESTIVE SYSTEM PROCESSES THE FOOD WE EAT.

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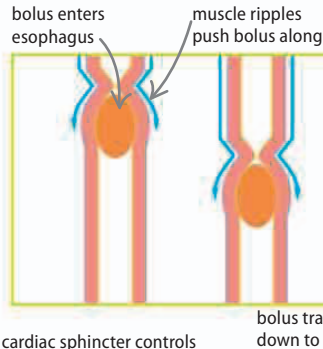
Digestion is a complex process that breaks down food into simple substances. These fats, sugars, proteins, and other nutrients are then absorbed, leaving unwanted waste to be expelled.

The digestive tract

Food is digested in the digestive tract—the passage food takes from the mouth to the anus. Nutrients are absorbed in the intestines (also known as the gut). The material that cannot be digested is mixed with other waste products from the body, such as the brown pigments from old blood cells, and pushed out of the body through the anus.

▷ Peristalsis

In the mouth, saliva is mixed with the food and it is chewed to form a bolus (ball). Muscles along the walls of the esophagus (throat) contract in waves to push the bolus down to the stomach.

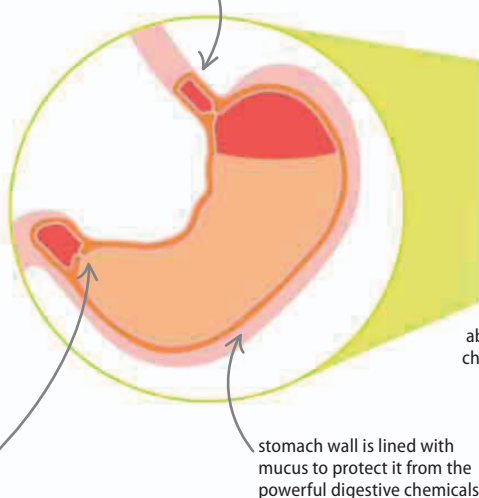


cardiac sphincter controls the opening of the stomach, stopping burning acids from leaking into the esophagus

▷ Stomach

The stomach is an elastic, muscular sac that can stretch to hold up to 4 liters (8 pints). The stomach churns up the food and mixes it with powerful acids and enzymes, turning it into a liquid. It is then sent to the small intestine, and from there to the large intestine.

food in the stomach passes through the pyloric sphincter to the small intestine after it has been thoroughly mixed with enzymes



senses from the nose and tastebuds on tongue give flavor to food

the tongue pushes food from the mouth to the pharynx

pharynx connects the nose and mouth to the oesophagus

salivary glands produce saliva, which begins digestion

esophagus

liver takes the products of digestion and processes them into substances useful for the body

gall bladder stores bile, a liquid used to break down fats

stomach

pancreas makes enzymes

duodenum

small intestine absorbs most of the chemicals from food

rectum stores waste until it can be ejected from the anus

anus

large intestine absorbs water from food